GATE ME 90 Days Challenge

Day 81/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Full Syllabus	Full-Length Mock 4	17 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on a mixed full-syllabus warm-up for the fourth full-length mock: antonyms, LCM timing, triangular determinants, dice probability, connected bodies over a pulley, compound bars, kinematic pairs, the simple pendulum, notch sensitivity, Reynolds number, cylindrical thermal resistance, constant-volume heating, the zeroth law, true strain and Little's law. Try them in **60 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 11 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 65 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Select the word that is most nearly OPPOSITE in meaning to TRANSIENT.

(A) fleeting
(B) momentary
(C) permanent
(D) temporary

Q2 NAT

Three alarms ring at intervals of 12, 15 and 20 minutes. They ring together at 8:00 a.m. How many times do they ring together from 8:00 a.m. to 2:00 p.m., both times included? _____

Q3 MCQ

The determinant of the matrix with rows (2, 0, 0), (3, −1, 0) and (4, 5, 3) is

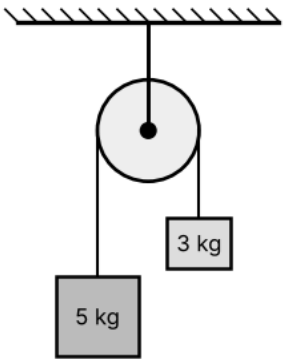
(A) 6
(B) −6
(C) 30
(D) −2

Q4 NAT

Two fair dice are thrown together. The probability that the sum of the numbers shown is at least 10 is _____ (round off to three decimal places)

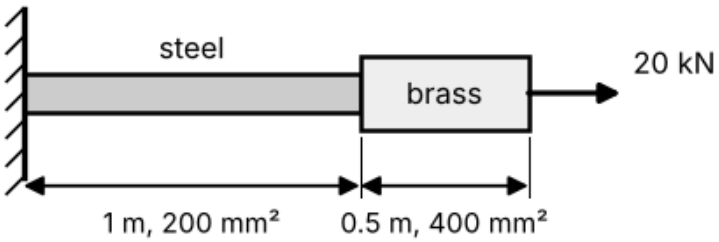
Q5 NAT

Two blocks of 5 kg and 3 kg hang from the ends of a light inextensible string passing over a light frictionless pulley, as shown. They are released from rest. Taking g = 9.81 m/s², the acceleration of the blocks (in m/s², round off to two decimal places) is _____



Q6 NAT

A compound bar is made of a steel rod (length 1 m, area 200 mm², E = 200 GPa) joined end to end with a brass rod (length 0.5 m, area 400 mm², E = 100 GPa), as shown. It carries an axial tensile load of 20 kN. The total elongation of the bar (in mm, round off to two decimal places) is _____



Q7 MCQ

A ball-and-socket joint (as in a car's steering linkage) is a

(A) spherical pair, allowing three independent rotations

(B) turning pair, allowing one rotation

(C) cylindrical pair, allowing one rotation and one sliding motion

(D) higher pair with point contact

Q8 NAT

The length of a simple pendulum whose period of small oscillation is 2 s at a place where g = 9.81 m/s² is _____ (in m, round off to three decimal places)

Q9 MCQ

A notched shaft has a theoretical stress concentration factor of 2.2, and the notch sensitivity of its material is 0.75. The fatigue stress concentration factor is

(A) 2.20

(B) 1.65

(C) 1.90

(D) 2.95

Q10 MCQ

Water (kinematic viscosity 1.0×10^{-6} m²/s) flows at a mean velocity of 0.04 m/s through a pipe of 50 mm diameter. The flow is

(A) laminar, with Re = 2000
(B) turbulent, with Re = 20 000
(C) laminar, with Re = 200
(D) turbulent, with Re = 2000

Q11 MCQ

A layer of insulation (k = 0.05 W/m·K) covers a pipe; its inner radius is 50 mm and its outer radius 100 mm. The conduction thermal resistance of a 1 m length of this insulation is about

(A) 1.00 K/W
(B) 2.21 K/W
(C) 0.32 K/W
(D) 4.41 K/W

Q12 NAT

2 kg of air (c_v = 0.718 kJ/kg·K) in a rigid closed tank is heated from 300 K to 500 K. The heat supplied (in kJ, round off to one decimal place) is _____

Q13 MCQ

The zeroth law of thermodynamics provides the basis for

(A) the definition of internal energy
(B) the concept of entropy
(C) the conservation of energy
(D) the measurement of temperature

Q14 NAT

A metal bar is stretched uniformly (before necking) from a gauge length of 100 mm to 150 mm. The true strain is _____ (round off to three decimal places)

Q15 MCQ

Customers arrive at a service counter at an average rate of 12 per hour and each spends on average 15 minutes in the system (waiting plus service). The average number of customers in the system is

(A) 12
(B) 0.8
(C) 3
(D) 15

Answer Key

Q	Type	Answer
Q1	MCQ	(C) permanent
Q2	NAT	7
Q3	MCQ	(B) −6
Q4	NAT	0.167 (accept 0.165 to 0.168)
Q5	NAT	2.45 (accept 2.43 to 2.47)
Q6	NAT	0.75 (accept 0.74 to 0.76)
Q7	MCQ	(A) spherical pair, allowing three independent rotations
Q8	NAT	0.994 (accept 0.988 to 1.000)
Q9	MCQ	(C) 1.90
Q10	MCQ	(A) laminar, with Re = 2000
Q11	MCQ	(B) 2.21 K/W
Q12	NAT	287.2 (accept 285.8 to 288.6)
Q13	MCQ	(D) the measurement of temperature
Q14	NAT	0.405 (accept 0.401 to 0.409)
Q15	MCQ	(C) 3

Detailed Solutions

Q1 **Answer: (C) permanent**

Concept: 'Transient' means lasting only a short time (as in a transient heat-conduction problem).

Fleeting, momentary and temporary all mean short-lived, so they are synonyms, not opposites.

The word meaning 'lasting for ever or for a long time' is **permanent**, option (C).

Exam tip: In antonym questions, first strike out every option that is a synonym — here three of the four were.

Q2 **Answer: 7**

Concept: Events with periods a, b, c coincide every LCM(a, b, c).

LCM(12, 15, 20) = 60 min, so they ring together every hour.

From 8:00 to 14:00 is 6 h, giving coincidences at 8, 9, 10, 11, 12, 1 and 2 o'clock: **7 times**.

Exam tip: Count both end points — 6 intervals contain 7 instants (the 'fence-post' trap answer is 6).

Q3 **Answer: (B) -6**

Concept: The determinant of a triangular matrix is the product of its diagonal entries.

The matrix is lower triangular (all entries above the diagonal are zero); its diagonal is 2, −1, 3.

det = 2 × (−1) × 3 = **−6**, option (B).

Exam tip: The same rule gives the eigenvalues of a triangular matrix directly: here 2, −1 and 3.

Q4 **Answer: 0.167 (accept 0.165 to 0.168)**

Concept: With two dice there are 36 equally likely ordered outcomes; count the favourable ones.

Sum 10: (4,6), (5,5), (6,4) — 3 ways. Sum 11: (5,6), (6,5) — 2 ways. Sum 12: (6,6) — 1 way.

P = 6/36 = 1/6 = **0.167**.

Exam tip: Count ordered pairs — (4,6) and (6,4) are different outcomes, but (5,5) is only one.

Q5 **Answer: 2.45 (accept 2.43 to 2.47)**

Concept: Write Newton's second law for each block with the same string tension T and the same magnitude of acceleration a.

5 kg (moves down): $5g - T = 5a$. 3 kg (moves up): $T - 3g = 3a$.

Adding: $2g = 8a$, so $a = 9.81/4 =$ **2.45 m/s²** (and $T = 3(g + a) = 36.8$ N).

Exam tip: $a = (m_1 - m_2)g/(m_1 + m_2)$ for an ideal pulley; the tension always lies between the two weights (29.4 N and 49.1 N).

Q6 **Answer: 0.75 (accept 0.74 to 0.76)**

Concept: Bars in series carry the same force; their elongations PL/(AE) add.

Steel: 20 000 × 1000/(200 × 200 000) = 0.50 mm. Brass: 20 000 × 500/(400 × 100 000) = 0.25 mm.

Total elongation = 0.50 + 0.25 = **0.75 mm**.

Exam tip: Series bars share the force and add deformations; parallel (side-by-side) bars share the deformation and add forces.

Q7 **Answer: (A) spherical pair, allowing three independent rotations**

Concept: Kinematic pairs are classified by the relative motions they permit (degrees of freedom).

A ball held in a spherical socket can rotate about any three perpendicular axes but cannot translate.

It is a **spherical pair with three rotational degrees of freedom**, option (A) — a lower pair, because the contact is over a surface.

Exam tip: Turning (revolute) and sliding (prismatic) pairs allow 1 DOF, a cylindrical pair 2 and a spherical pair 3; point or line contact makes a higher pair.

Q8 **Answer: 0.994 (accept 0.988 to 1.000)**

Concept: For small oscillations of a simple pendulum, $T = 2\pi\sqrt{(L/g)}$.

$L = gT^2/(4\pi^2) = 9.81 \times 4/(4 \times 9.8696)$.

$L = 9.81/9.8696 =$ **0.994 m**.

Exam tip: The period is independent of the bob's mass and (for small swings) of the amplitude — only L and g matter.

Q9 **Answer: (C) 1.90**

Concept: $K_f = 1 + q(K_t - 1)$, where q (0 to 1) is the notch sensitivity.

$K_t - 1 = 1.2$; $q(K_t - 1) = 0.75 \times 1.2 = 0.9$.

$K_f = 1 + 0.9 =$ **1.90**, option (C).

Exam tip: q = 0 means no notch effect ($K_f = 1$) and q = 1 means full effect ($K_f = K_t$); 1.65 is the wrong product $q \times K_t$.

Q10 **Answer: (A) laminar, with Re = 2000**

Concept: Re = VD/ν; pipe flow is laminar below about 2000–2300.

Re = $0.04 \times 0.05/(1.0 \times 10^{-6}) = 2000$.

Re = 2000 is in the laminar range, so the flow is **laminar, with Re = 2000**, option (A).

Exam tip: Convert the diameter to metres first — 50 instead of 0.05 shifts Re by a factor of 1000.

Q11 **Answer: (B) 2.21 K/W**

Concept: For radial conduction through a cylindrical layer, $R = \ln(r_2/r_1)/(2\pi kL)$.

$\ln(100/50) = \ln 2 = 0.693$; $2\pi kL = 2\pi \times 0.05 \times 1 = 0.3142$ W/K.

R = 0.693/0.3142 = **2.21 K/W**, option (B).

Exam tip: Treating the layer as a plane wall of thickness 50 mm with the inner area ($2\pi \times 0.05 \times 1$) gives 3.18 K/W; with the outer area 1.59 K/W — the log form lies between.

Q12 **Answer: 287.2 (accept 285.8 to 288.6)**

Concept: In a rigid (constant-volume) closed system no boundary work is done, so $Q = \Delta U = mc_v\Delta T$.

$\Delta T = 500 - 300 = 200$ K.

$Q = 2 \times 0.718 \times 200 =$ **287.2 kJ**.

Exam tip: Using c_p (1.005) gives 402 kJ — that is the constant-pressure answer, where the extra heat goes into boundary work.

Q13 **Answer: (D) the measurement of temperature**

Concept: Zeroth law: two bodies each in thermal equilibrium with a third body are in thermal equilibrium with each other.

So a thermometer (the 'third body') can compare the temperatures of bodies without their touching.

This is the basis of **temperature measurement**, option (D).

Exam tip: First law → energy and internal energy; second law → entropy and the limits on efficiency.

Q14 **Answer: 0.405 (accept 0.401 to 0.409)**

Concept: True strain is the integral of dL/L: $\varepsilon_T = \ln(L/L_0) = \ln(1 + e)$, where e is the engineering strain.

Engineering strain e = (150 − 100)/100 = 0.5.

$\varepsilon_T = \ln 1.5 =$ **0.405**.

Exam tip: True strains add for successive steps ($\ln 1.2 + \ln 1.25 = \ln 1.5$), engineering strains do not; for small strains the two are nearly equal.

Q15 **Answer: (C) 3**

Concept: Little's law: L = λW, valid for any stable queue in steady state.

λ = 12 per hour; W = 15 min = 0.25 h.

L = 12 × 0.25 = **3**, option (C).

Exam tip: Keep λ and W in the same time unit; using W = 15 (minutes) with λ per hour gives the absurd 180.

What next?

Day 81 test tonight, 7 to 11 PM

65 questions · 180 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

Dig Career Thrust app

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Next, Day 82/90: Formula Sprint: Thermal. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.